

Supplementary material

Appendix I: Survey documentation

Adherence to the Principles and Guidance for Human Subjects Research

The survey experiment detailed in the paper received full IRB approval (University of Geneva, CUREG-2024-07-87). Participants were recruited via the online survey panel from bilendi and programmed in Qualtrics. Bilendi handles incentivising the participants and the researchers pay bilendi a cost per completed interview (CPI) and bilendi provides a portion of those earnings to participants. Before entering the survey, participants first completed a consent form that included the researcher's name and affiliation, the general purpose of the research, an explanation of the survey (including duration), the costs and benefits of participation, and contact information for the lead researcher. They were also informed that no identifying information would be collected, and that they were able to opt out of the research at any time. The project did not involve any deception, and no identifying information about the participants was collected. We also did not intervene in any political processes.

Pre-registration

This study was pre-registered on the Open Science Framework (OSF) before starting the data collection [https://osf.io/7x9nm/overview?view_only=50230b81235849889502c9799d6a66fc]. The pre-registration contains the two measurement approaches of responsibility-sharing preferences, the item battery and the conjoint experiment, but no specific hypotheses as the study focuses on exploring the feasibility of multi-dimensional responsibility-sharing.

Sampling

For each country, we collect a sample of N=1,700 respondents from the adult resident population (total sample size of N=10,200), with country-specific quotas for age (18-24, 25-34, 35-44, 45-54, 55-79), gender (male, female) and education (low, high). We included a commitment check at the beginning of the survey and an attention check at the end of the survey. Respondents who failed either were excluded.

Questionnaire

The survey items are part of a larger questionnaire on public attitudes and preferences on migration and asylum governance in the European Union. The survey was fielded in the six national languages of the country sample (German, Danish, Italian, Greek, Polish, and Hungarian), first professionally translated by the survey company bilendi and then double-checked with native speakers for clarity and comprehension. There are three parts: a first block of socio-demographic questions and general political attitudes, then a block on specific preferences on responsibility-sharing, and finally the conjoint experiment.

PART I: General variables and attitudes

Age

First, we would like to learn a little about you as a person. How old are you?

Gender

Please indicate your gender.

Response scale: Male; Female; Other

Education

What is your highest level of education?

Response scale: Primary school not completed; Primary school certificate; Secondary school certificate, internship, vocational preparation year; A-levels, technical college/university entrance qualification, completed apprenticeship, college; Technical secondary school, technical college, vocational school, evening school, dual system; Technical/vocational academy, administrative college, technical college; Bachelor's degree, state-certified technician, master craftsman's certificate, diploma from a technical college; Master's degree, diploma from a university

Left-right orientation

In politics, people often talk about 'left' and 'right'. Where would you place yourself on this scale?

Response scale: 0 (Left) to 10 (Right)

Euroscepticism

Some say European unification should be pushed further. Others say it already has gone too far. What is your opinion? Please indicate your views using a scale from 0 to 10, where '0' means unification "has already gone too far" and '10' means it "should be pushed further".

Immigration attitudes

Is [COUNTRY] made a worse or a better place to live by people coming to live here from other countries?

Response scale: 0 Worse place to live to 10 Better place to live

Refugee attitudes

Below we ask you a few questions about what you think about the political treatment of refugees. Refugees are people who have to flee their country for fear of political persecution and need protection in a safe country.

What is your opinion on the following statements?

Refugees in need of protection should be given humanitarian protection in a safe country.

Response scale: Do not agree at all (1) Rather disagree (8) Partly disagree (2) Neutral (4) Partly agree (5) Rather agree (6) Completely agree (7)

PART II: Preferences on responsibility-sharing

Intro responsibility-sharing

Below we ask you a few questions about what you think about the political treatment of refugees. Refugees are people who are forced to flee their country for fear of political persecution and need protection in a safe country.

There are various ways in which [country] can contribute to the protection of refugees.

To what extent do you agree with the following contributions?

Norms (EU)

Standardised European rules for deciding on asylum applications, laid down by the European Union.

Response scale: 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

Norms (international)

Standardised international rules for deciding on asylum applications, established by the United Nations.

Response scale: 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

Money (EU)

Financial support for other EU countries to relieve them of the burden of taking in refugees.

Response scale: 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

Money (international)

Financial support to countries outside the European Union to help them take in refugees.

Response scale: 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

People (EU)

Acceptance of refugees from other EU countries in order to relieve them of the burden of taking in refugees.

Response scale: 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

People (international)

Accepting refugees from countries outside the European Union in order to relieve the burden on these countries when accepting refugees.

Response scale: 1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

The next part of the survey is dedicated to migration control. European countries are endeavouring to control migration and protect the EU's external borders. There are various ways in which [country] can contribute to migration control.

To what extent do you agree with the following contributions?

Control (EU)

Financial and personnel support for the European border protection agency Frontex to combat illegal migration.

1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

Control (international)

Financial support for third countries outside Europe to combat illegal migration to Europe.

1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

PART III: Conjoint experiment

Design: Comparison of policy-pairs (two iterations), full randomization of the levels in each factor)

Below you can see two models of how Europe could implement a common migration policy. We will then ask you for your opinion on these proposals.

Model A

Norms/Money/People *Refugee protection*
EU/International *Migration control*
Mandatory/Voluntary *Bindingness*

Model B

Norms/Money/People
EU/International
Mandatory/Voluntary

Refugee protection

- Cost sharing: States with few refugees support states with many refugees through financial solidarity. [money-sharing]
- Relocation: States with few refugees take in refugees from states with many refugees. [people-sharing]
- Harmonization: Uniform standards for assessing asylum applications within the EU. [norm-sharing]

Migration control

- Joint protection of the European external borders by the member states. [EU control]
- European support for non-European countries so that they prevent irregular migration to Europe. [Int. control]

Bindingness

- Voluntary: Each member state freely decides whether to participate in the common policy.
- Mandatory: Member states are obliged to participate in the common policy.

Which of the two models of European migration policy do you favour? Model A or Model B

Model A; Model B

How would you rate Model A?

1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

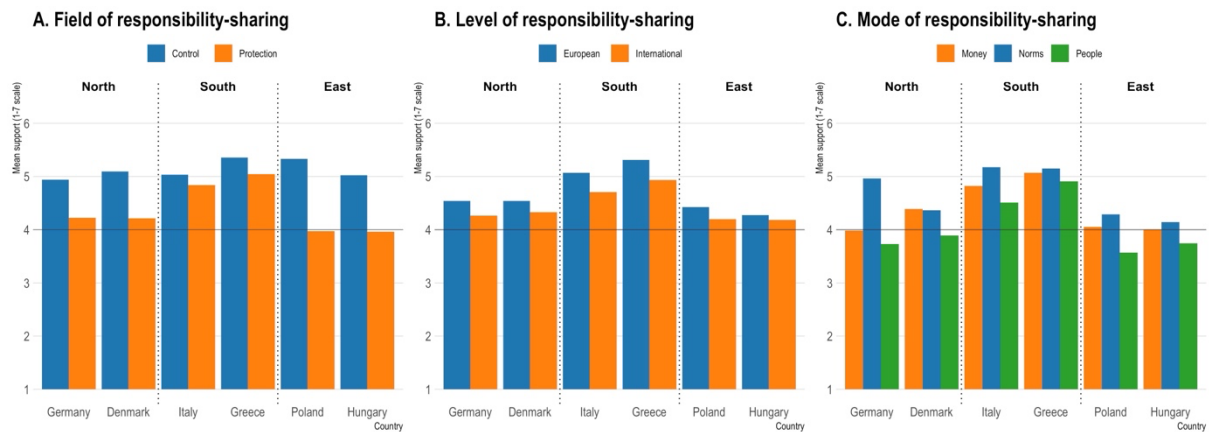
How would you rate Model B?

1 Strongly disagree; 2 Disagree; 3 Somewhat disagree; 4 Neither agree nor disagree; 5 Somewhat agree; 6 Agree; 7 Strongly agree

→ conjoint experiment with two iterations (evaluation of Model A & B as well as C & D)

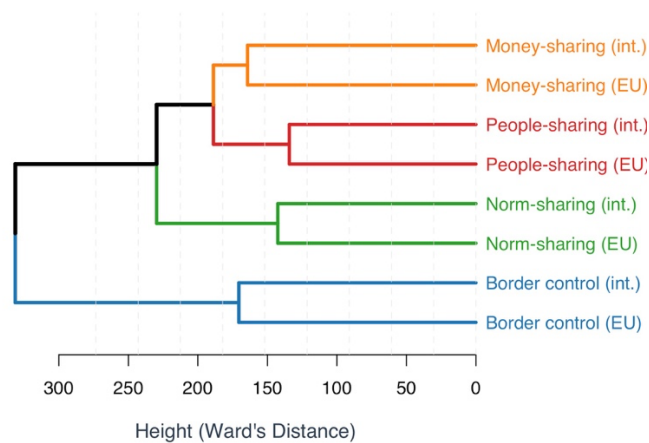
Appendix II: Additional analyses

Figure A1: Mean support for different contributions to responsibility-sharing



Note: The bar plots display the mean support for different responsibility-sharing contributions based on the mean of the support items with a seven-point Likert scale (including neutral responses). The indicators are aggregated based on three different dimensions: the field, level and mode of responsibility-sharing.

Figure A2: Dendrogram of responsibility-sharing preferences



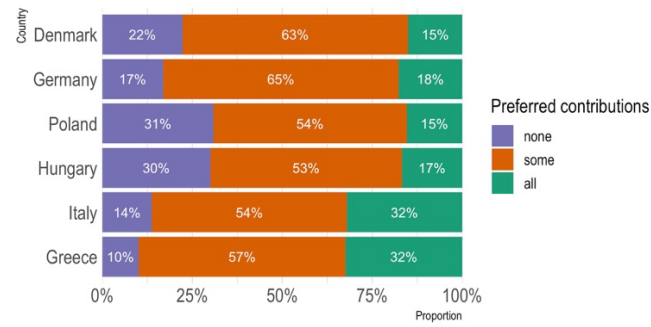
Note: Dendrogram of eight immigration-related variables based on their pairwise Euclidean distances. Clustering was performed using Ward's method on standardised variable scores. The x-axis reflects the dissimilarity (distance) at which variables (or variable groups) are joined. Labels are grouped into four colour-coded clusters, reflecting thematic similarity in how variables co-occur across cases.

Table A1: Factor loadings for responsibility-sharing preferences

	ML1	ML2	ML3	ML4
Norm-sharing (EU)	-0.03	0.82	-0.02	0.01
Norm-sharing (int.)	0.08	0.77	0.02	-0.01
Money-sharing (EU)	0.01	0.02	-0.01	0.98
Money-sharing (int.)	0.42	-0.02	0.21	0.31
People-sharing (EU)	0.66	0.09	-0.02	0.16
People-sharing (int.)	0.88	0.04	0.00	-0.04
Border control (EU)	-0.23	0.15	0.49	0.10
Border control (int.)	0.02	-0.02	1.00	-0.02

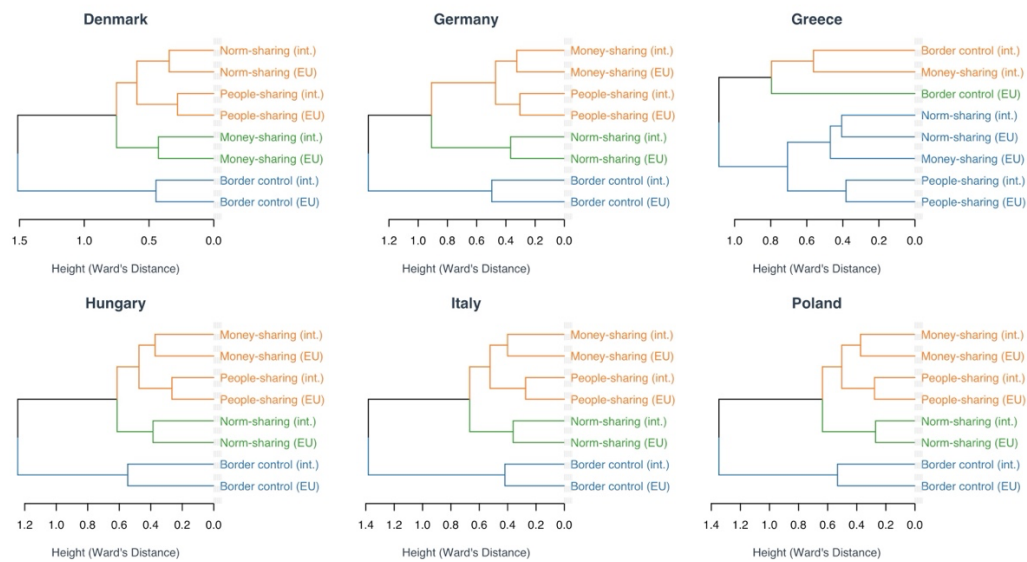
Note: Factor analysis using maximum likelihood extraction and oblimin rotation. The two highest loadings per factor are highlighted.

Figure A3: Principled and conditional support for refugee protection contributions



Note: The same bar plot as Figure 2 in the manuscript, but data confined to contributions to refugee protection.

Figure A4: Dendrograms by country

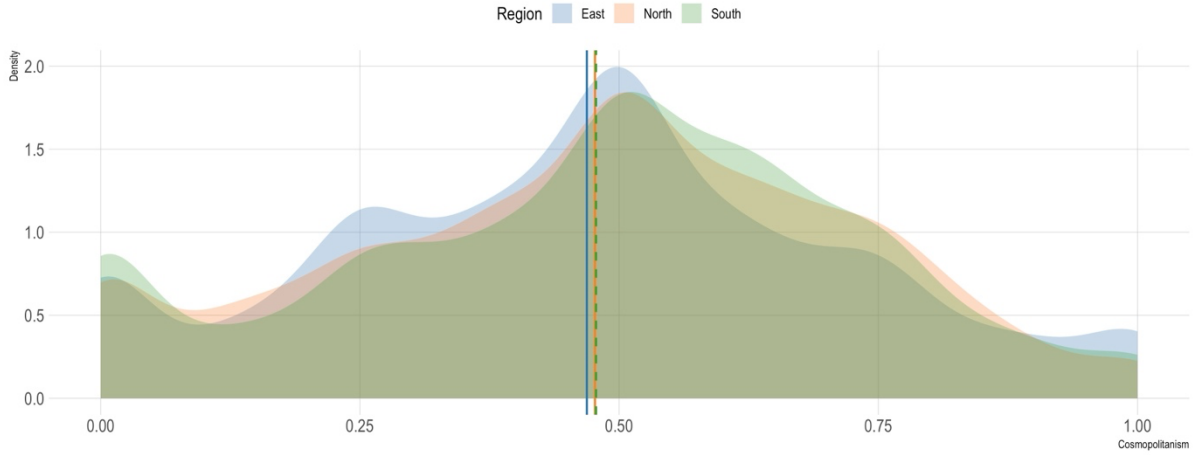


Note: Dendrograms of eight immigration-related variables based on their pairwise Euclidean distances separately for the six countries in the sample. Clustering was performed using Ward's method on standardized variable scores. The x-axis reflects the dissimilarity (distance) at which variables (or variable groups) are joined.

Figure A5: Correlation plot responsibility-sharing preferences



Figure A6: Density plot of political ideology by region



Note: Overlapping and semi-transparent density plot showing the distribution of cosmopolitanism (vs. communitarianism) in Eastern Europe (Poland, Hungary), Northern Europe (Germany, Denmark), and Southern Europe (Italy, Greece). Colored vertical lines show the regional means.

Table A2: Multilevel model estimates

	DV: Support responsibility-sharing contribution						
	Policy field		Governance level		Contribution mode		
	Refugee protection (1)	Migration control (2)	European Union (3)	Inter-national (4)	Norm sharing (5)	Money sharing (6)	People sharing (7)
East	-0.04	0.03	-0.03	-0.02	-0.07	-0.02	-0.02
(vs North)	(0.02)	(0.03)	(0.01)	(0.02)	(0.04)	(0.04)	(0.04)
South	0.12***	0.03	0.11***	0.09***	0.08*	0.13***	0.15***
(vs North)	(0.02)	(0.03)	(0.01)	(0.02)	(0.04)	(0.04)	(0.04)
Cosmopolitan	0.45***	-0.01	0.34***	0.34***	0.44***	0.41***	0.51***
(vs. Communitarian)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Constant	0.32***	0.68***	0.43***	0.39***	0.40***	0.34***	0.23***
	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.03)	(0.03)
Observations	10,200	10,200	10,200	10,200	10,200	10,200	10,200
Log Likelihood	2,491.64	624.27	3,321.84	2,777.58	686.97	237.66	469.53
AIC	-4,971.27	-1,236.55	-6,631.67	-5,543.15	-1,361.93	-463.33	-927.06
BIC	-4,927.89	-1,193.17	-6,588.29	-5,499.77	-1,318.55	-419.95	-883.68

Note: Multilevel models with random intercepts. North serves as the reference category for regional dummies. Cosmopolitan is an ideological variable capturing the extent of cosmopolitanism (vs. communitarianism), based on pro-EU and pro-immigration attitudes. All dependent and independent variables are rescaled to range from 0 to 1 to facilitate interpretation and comparison. Coefficients rounded to two decimal places. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table A3: Multilevel model estimates with control variables

<i>DV: Support responsibility-sharing contribution</i>							
	Policy field		Governance level		Contribution mode		
	Refugee protection	Migration control	European Union	Inter-national	Norm sharing	Money sharing	People sharing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
East	-0.04	0.03	-0.03	-0.01	-0.07	-0.02	-0.02
(vs. North)	(0.02)	(0.03)	(0.01)	(0.02)	(0.04)	(0.04)	(0.04)
South	0.12***	0.03	0.11***	0.09***	0.09*	0.13***	0.15***
(vs. North)	(0.02)	(0.03)	(0.01)	(0.02)	(0.04)	(0.04)	(0.04)
Cosmopolitan	0.45***	-0.01	0.33***	0.33***	0.43***	0.41***	0.51***
(vs. Communitarian)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
<i>Control variables:</i>							
Age (years)	0.001***	0.002***	0.001***	0.001***	0.001***	0.001***	-0.0003*
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0001)
Gender (male)	-0.0003	0.03***	0.01*	0.01*	-0.01*	0.02***	-0.01
	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.005)	(0.005)
Gender (other)	0.04	-0.15*	0.01	-0.02	0.06	0.03	0.03
	(0.05)	(0.06)	(0.04)	(0.05)	(0.06)	(0.06)	(0.06)
Education (low)	-0.03***	-0.02***	-0.03***	-0.02***	-0.04***	-0.03***	-0.01
	(0.004)	(0.01)	(0.004)	(0.004)	(0.01)	(0.01)	(0.01)
Constant	0.31***	0.57***	0.39***	0.35***	0.38***	0.30***	0.25***
	(0.02)	(0.03)	(0.01)	(0.02)	(0.03)	(0.03)	(0.03)
Observations	10,174	10,174	10,174	10,174	10,174	10,174	10,174
Log Likelihood	2,501.10	761.26	3,371.34	2,804.75	737.32	273.34	457.45
AIC	-4,982.19	-1,502.52	-6,722.68	-5,589.50	-1,454.64	-526.68	-894.90
BIC	-4,909.92	-1,430.24	-6,650.40	-5,517.23	-1,382.37	-454.41	-822.62

*Note: Multilevel models with random intercepts. North serves as the reference category for regional dummies. Cosmopolitan is an ideological variable capturing the extent of cosmopolitanism (vs. communitarianism), based on pro-EU and pro-immigration attitudes. All dependent and independent variables are rescaled to range from 0 to 1 to facilitate interpretation and comparison. Coefficients rounded to two decimal places. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Table A4: Multilevel model estimates with alternative ideology measure

<i>DV: Support responsibility-sharing contribution</i>							
	Policy field		Governance level		Contribution mode		
	Refugee protection	Migration control	European Union	Inter-national	Norm sharing	Money sharing	People sharing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
East	-0.03*	0.02	-0.03	-0.01	-0.07	-0.02	-0.02
(vs. North)	(0.01)	(0.03)	(0.02)	(0.02)	(0.04)	(0.03)	(0.03)
South	0.12***	0.03	0.11***	0.09***	0.09*	0.13***	0.15***
(vs. North)	(0.01)	(0.03)	(0.02)	(0.02)	(0.04)	(0.03)	(0.03)
Right-wing	-0.19***	0.13***	-0.12***	-0.10***	-0.22***	-0.14***	-0.22***
(vs. left-wing)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Constant	0.64***	0.60***	0.65***	0.60***	0.72***	0.60***	0.58***
	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.02)	(0.02)
Observations	10,200	10,200	10,200	10,200	10,200	10,200	10,200
Log Likelihood	1,172.87	727.50	2,372.46	1,892.14	-165.91	-558.78	-648.50
AIC	-2,333.74	-1,443.01	-4,732.93	-3,772.27	343.83	1,129.56	1,309.00
BIC	-2,290.36	-1,399.63	-4,689.55	-3,728.89	387.21	1,172.94	1,352.38

*Note: Robustness check using models from Table A2, replacing the cosmopolitanism-communitarianism scale with a left-right scale as an alternative ideology measure. All variables are rescaled to range from 0 to 1 to facilitate interpretation and comparison. Coefficients rounded to two decimal places. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Table A5: Multilevel model estimates with interaction terms

<i>DV: Support responsibility-sharing contribution</i>							
	Policy field		Governance level		Contribution mode		
	Refugee protection	Migration control	European Union	Inter-national	Norm sharing	Money sharing	People sharing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
East (vs. North)	-0.01 (0.02)	-0.02 (0.03)	-0.02 (0.02)	-0.02 (0.02)	-0.08* (0.04)	0.002 (0.04)	0.03 (0.05)
South (vs. North)	0.22*** (0.03)	0.05 (0.03)	0.19*** (0.02)	0.16*** (0.02)	0.15*** (0.04)	0.23*** (0.04)	0.27*** (0.05)
Cosmopolitan (vs. Communitarian)	0.54*** (0.01)	-0.03* (0.02)	0.41*** (0.01)	0.38*** (0.01)	0.48*** (0.02)	0.50*** (0.02)	0.63*** (0.02)
East*Cosmopolitan	-0.05** (0.02)	0.10*** (0.02)	-0.03 (0.02)	0.01 (0.02)	0.02 (0.02)	-0.05* (0.02)	-0.12*** (0.02)
South*Cosmopolitan	-0.20*** (0.02)	-0.04 (0.02)	-0.17*** (0.02)	-0.14*** (0.02)	-0.14*** (0.02)	-0.22*** (0.02)	-0.25*** (0.02)
Constant	0.28*** (0.02)	0.69*** (0.02)	0.40*** (0.01)	0.37*** (0.02)	0.38*** (0.03)	0.29*** (0.03)	0.17*** (0.03)
Observations	10,200	10,200	10,200	10,200	10,200	10,200	10,200
Log Likelihood	2,551.50	640.45	3,378.23	2,817.46	712.57	280.73	527.44
AIC	-5,086.99	-1,264.90	-6,740.46	-5,618.93	-1,409.13	-545.46	-1,038.87
BIC	-5,029.15	-1,207.06	-6,682.62	-5,561.09	-1,351.29	-487.62	-981.03

Note: Multilevel models with random intercepts. North serves as the reference category for regional dummies. Cosmopolitan is an ideological variable capturing the extent of cosmopolitanism (vs. communitarianism), based on pro-EU and pro-immigration attitudes. All variables are rescaled to range from 0 to 1 to facilitate interpretation and comparison. Coefficients rounded to two decimal places. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table A6: OLS regression model estimates

<i>DV: Support responsibility-sharing contribution</i>							
	Policy field		Governance level		Contribution mode		
	Refugee protection	Migration control	European Union	Inter-national	Norm sharing	Money sharing	People sharing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
East (vs. North)	-0.04 (0.02)	0.03 (0.03)	-0.03*** (0.01)	-0.02 (0.02)	-0.07 (0.05)	-0.02 (0.04)	-0.02 (0.05)
South (vs. North)	0.12*** (0.01)	0.03 (0.03)	0.11*** (0.02)	0.09*** (0.02)	0.08 (0.05)	0.13** (0.04)	0.15*** (0.03)
Cosmopolitan (vs. Communitarian)	0.45*** (0.04)	-0.01 (0.04)	0.34*** (0.04)	0.33*** (0.03)	0.44*** (0.03)	0.41*** (0.05)	0.50*** (0.05)
Constant	0.32*** (0.02)	0.67*** (0.03)	0.43*** (0.02)	0.39*** (0.02)	0.40*** (0.05)	0.34*** (0.06)	0.23*** (0.03)
Observations	10,200	10,200	10,200	10,200	10,200	10,200	10,200
R ²	0.33	0.004	0.26	0.21	0.24	0.21	0.29
Adjusted R ²	0.33	0.003	0.26	0.21	0.24	0.21	0.29

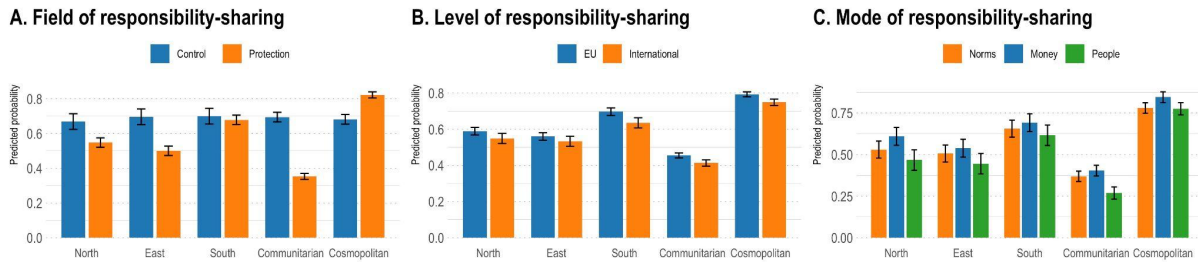
Note: OLS regression models with country-clustered standard errors. North serves as the reference category for regional dummies. Cosmopolitan is an ideological variable capturing the extent of cosmopolitanism (vs. communitarianism), based on pro-EU and pro-immigration attitudes. All dependent variables are rescaled to range from 0 to 1 to facilitate interpretation and comparison. Coefficients rounded to two decimal places. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table A7: Estimates of conjoint experiment

DV: Model choice (1-7 scale)	
Refugee protection: Norms (vs Money)	0.053** (0.016)
Refugee protection: People (vs Money)	-0.067*** (0.016)
Border control: International (vs European)	-0.113*** (0.013)
Bindingness: Voluntary (vs Mandatory)	-0.050*** (0.013)
Germany (vs Denmark)	-0.013 (0.027)
Greece (vs Denmark)	0.110*** (0.027)
Hungary (vs Denmark)	-0.098*** (0.027)
Italy (vs Denmark)	0.035 (0.027)
Poland (vs Denmark)	-0.099*** (0.027)
Constant	4.237*** (0.023)
Observations	40,800
Log Likelihood	-70,636.090
AIC	141,296.200
BIC	141,399.600

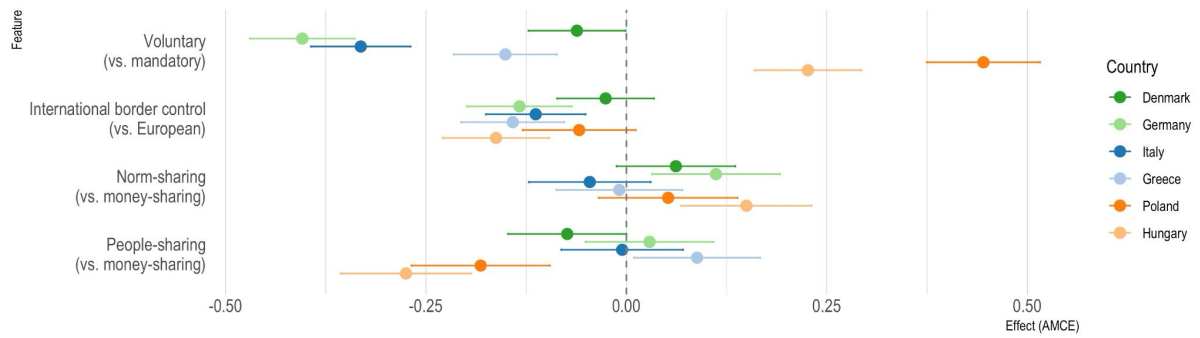
*Note: Results from a linear mixed-effects model with respondent-level random intercepts and country fixed effects. The dependent variable is the attribute choice rating. Attribute levels are coded relative to their respective baseline categories (Money, National border, Mandatory rules, Denmark). Levels of statistical significance are indicated as follows: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Figure A7: Predicted support of responsibility-sharing contributions by region and ideology



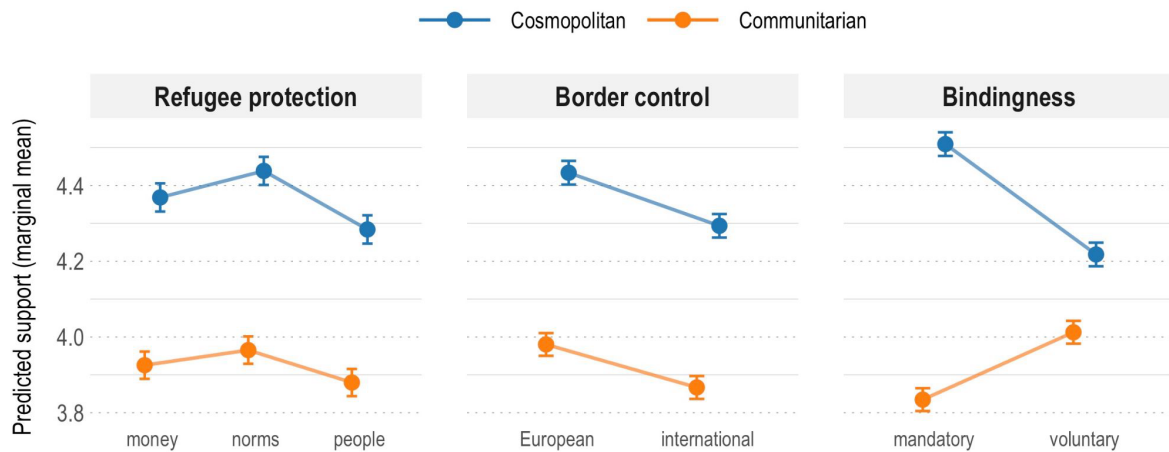
Note: Predicted probabilities of different responsibility-sharing contributions based on a linear mixed-effects model with country-level random intercepts. All dependent variables scaled to a range of 0 to 1 to facilitate interpretation and comparison. Error bars represent 95% confidence intervals.

Figure A8: Predicted probabilities of conjoint experiment by country



Note: Average Marginal Component Effects (AMCEs) from conjoint experiment with separate estimates by country. Points indicate effect sizes relative to reference categories (in brackets); horizontal bars represent 95% confidence intervals.

Figure A9: Predicted probabilities of conjoint experiment by political ideology



Note: Coefficient plots based on multilevel models showing predicted support for responsibility-sharing dimensions for separate samples of Cosmopolitans ($N=25,650$) and Communitarians ($N=27,432$). Those respondents exactly in the middle of the ideological scale are excluded. Point estimates with 95% confidence intervals shown.